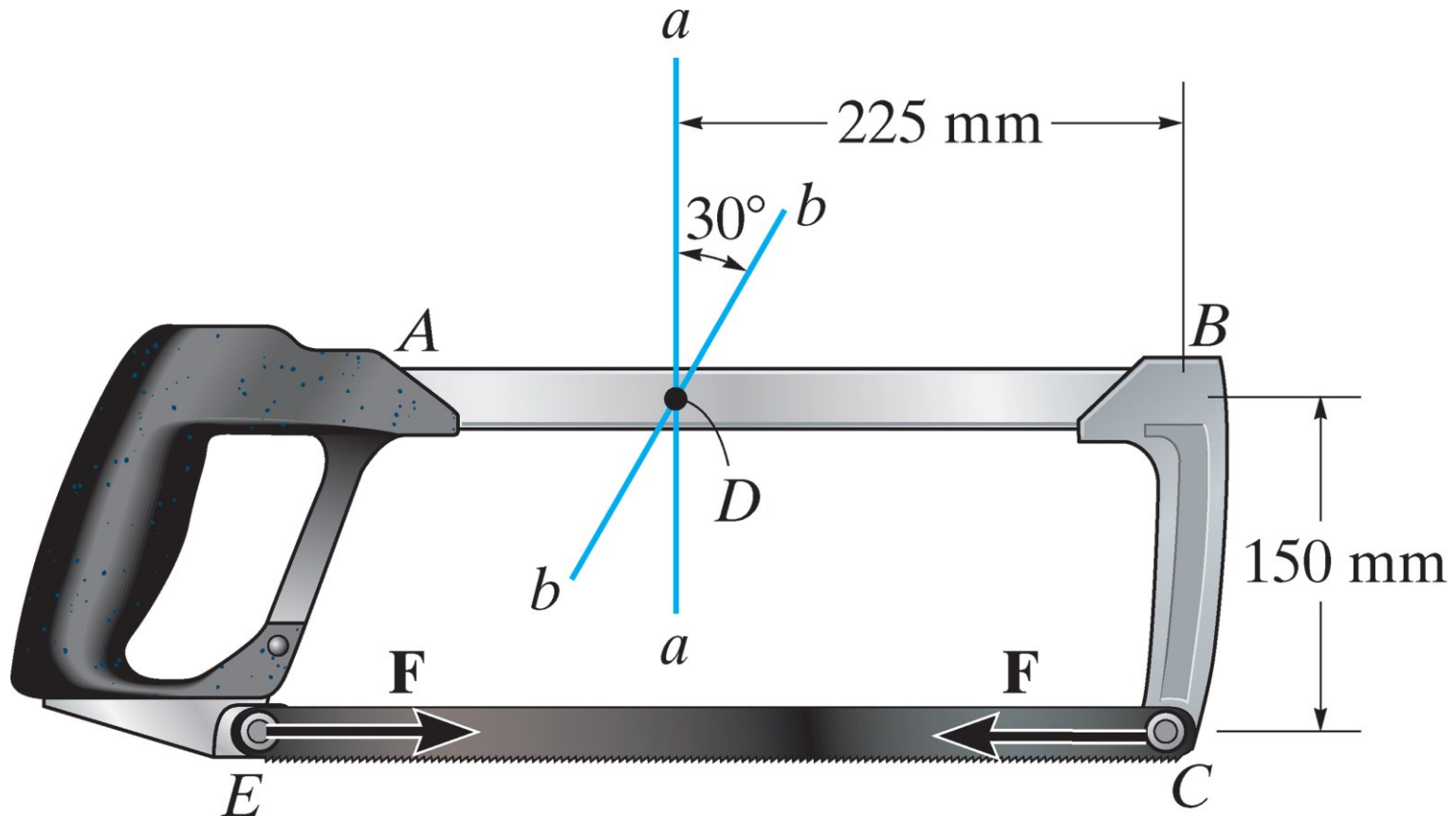


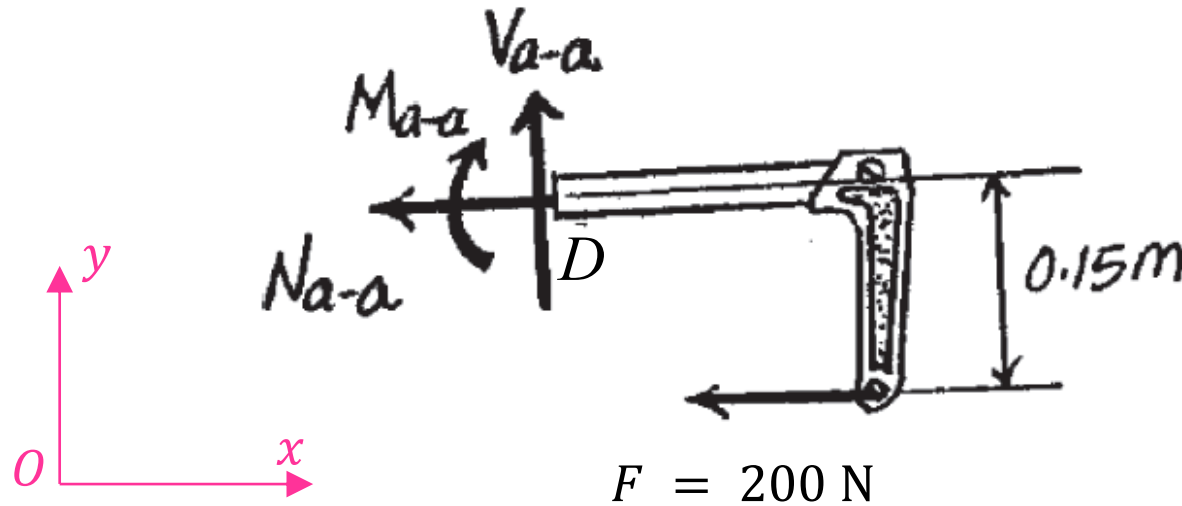
1. The blade of the hacksaw is subjected to a pretension force of $F = 200\text{ N}$. Determine the resultant internal loadings acting on section $a-a$ and $b-b$ that passes through point D , respectively. (20' points)



Solutions to Assignment #1:

For section *a-a*:

Internal Loadings: Referring to the free-body diagram of the section of the hacksaw,



$$\sum F_x = 0 \quad \rightarrow \quad N_{a-a} + 200 = 0 \quad \rightarrow \quad N_{a-a} = -200 \text{ N (Ans.)}$$

$$\sum F_y = 0 \quad \rightarrow \quad V_{a-a} = 0 \quad \text{(Ans.)}$$

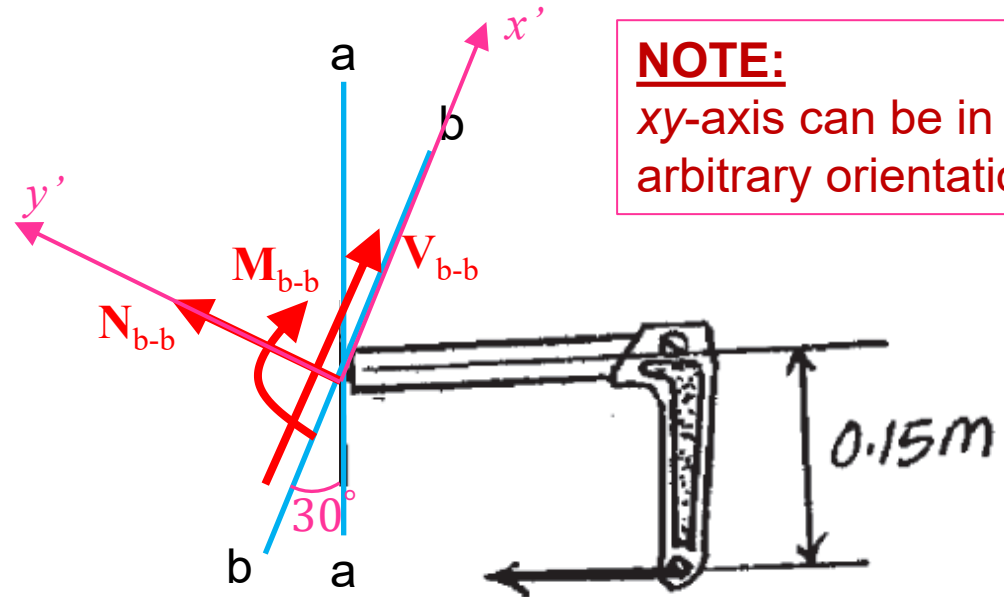
$$\sum M_D = 0 \quad \rightarrow \quad M_{a-a} + 200 \times 0.15 = 0 \quad \rightarrow \quad M_{a-a} = -30 \text{ N} \cdot \text{m (Ans.)}$$

The negative sign indicates that N_{a-a} and M_{a-a} act in the opposite sense to that shown on the free-body diagram.

Solutions to Assignment #1, cont'd:

For section *b-b*:

Internal Loadings: In the previous free-body diagram of the section of the hacksaw, rotate the normal force and shear force by clock-wise 30° , and then apply the force and moment equations of equilibrium:



$$F = 200 \text{ N}$$

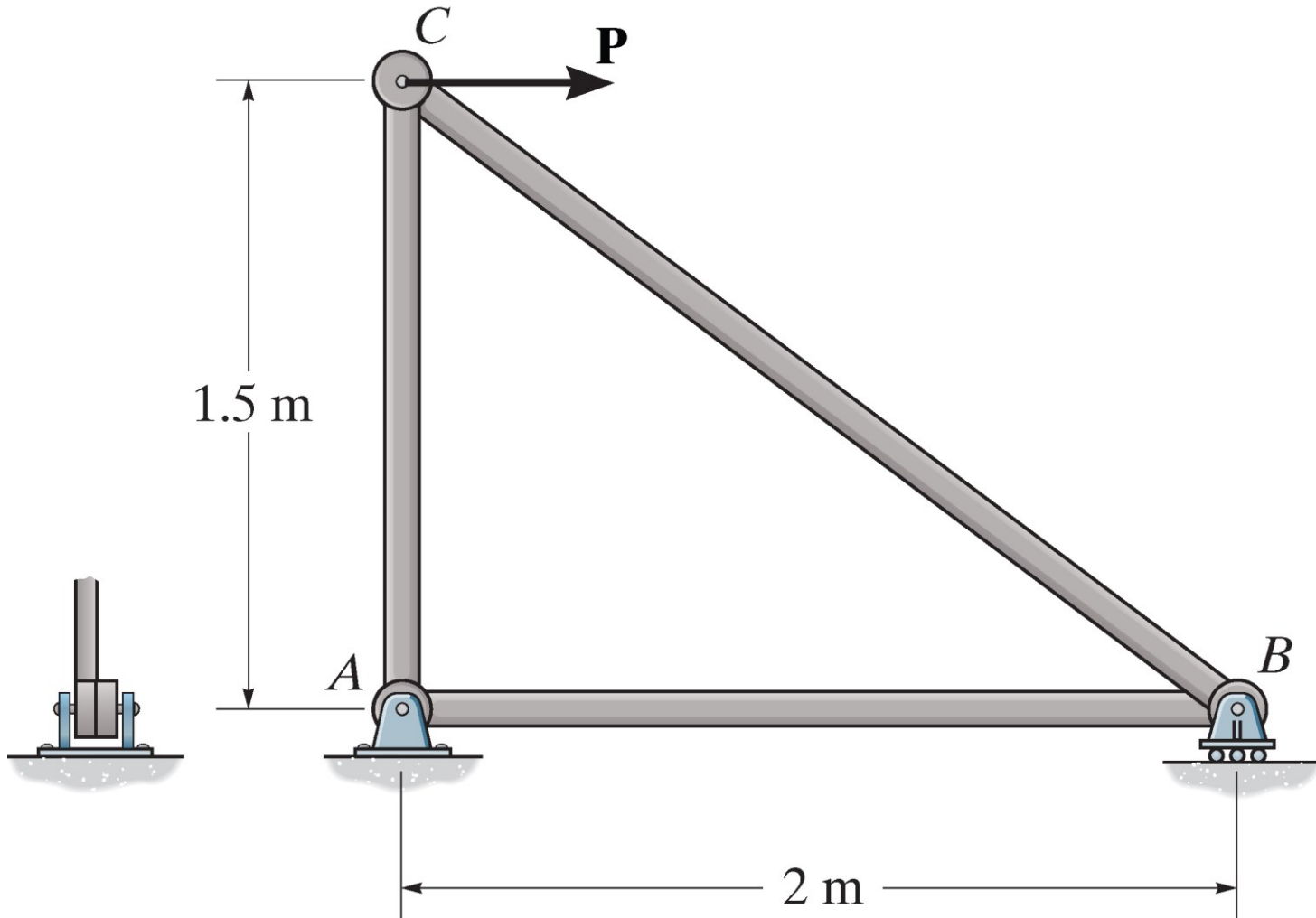
$$\sum F_{x'} = 0 \quad \rightarrow \quad V_{b-b} - 200 \sin 30^\circ = 0 \quad \rightarrow \quad V_{b-b} = 100 \text{ N} \quad (\text{Ans.})$$

$$\sum F_{y'} = 0 \quad \rightarrow \quad N_{b-b} + 200 \cos 30^\circ = 0 \quad \rightarrow \quad N_{b-b} = -173.2 \text{ N} \quad (\text{Ans.})$$

$$\sum M_D = 0 \quad \rightarrow \quad M_{b-b} + 200 \times 0.15 = 0 \quad \rightarrow \quad M_{b-b} = -30 \text{ N} \cdot \text{m} \quad (\text{Ans.})$$

The negative sign indicates that N_{b-b} and M_{b-b} act in the opposite sense to that shown on the free-body diagram.

2. Determine the average normal stress in each of the 20-mm-diameter bars of the truss. Set $P = 40$ kN. (15' points)



TIPS: first solve the statics problem, then solve the deformation problem

Solutions to Assignment #2:

Internal Loadings: First, consider the equilibrium of joint C:

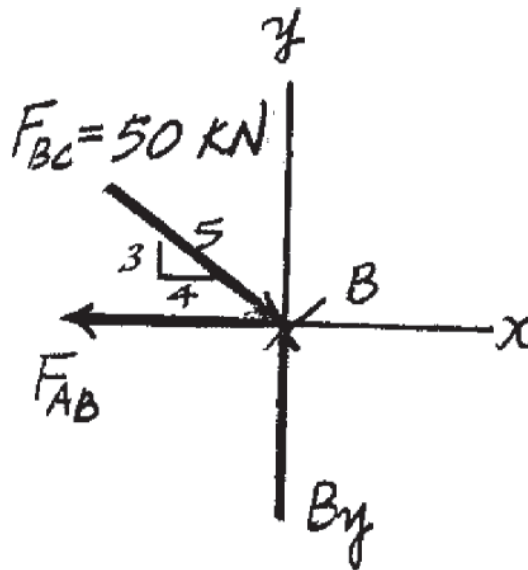
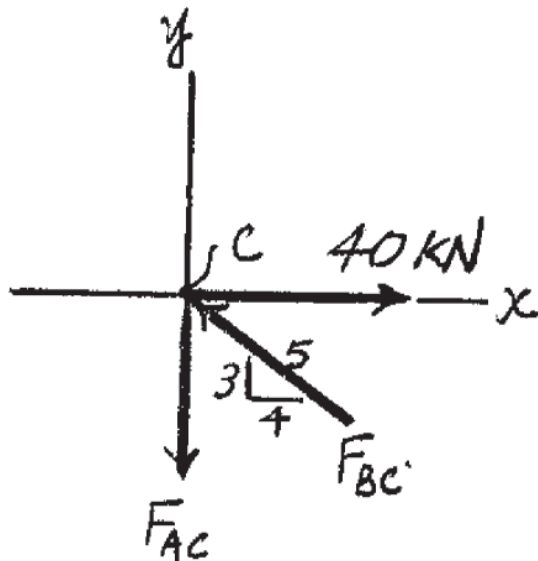
$$\begin{aligned} \pm \rightarrow \Sigma F_x = 0; \qquad 40 - F_{BC} \left(\frac{4}{5} \right) &= 0 & F_{BC} = 50 \text{ kN (C)} \end{aligned}$$

$$\begin{aligned} + \uparrow \Sigma F_y = 0; \qquad 50 \left(\frac{3}{5} \right) - F_{AC} &= 0 & F_{AC} = 30 \text{ kN (T)} \end{aligned}$$

Then, for joint B, apply equation of equilibrium:

$$\pm \rightarrow \Sigma F_x = 0; \qquad 50 \left(\frac{4}{5} \right) - F_{AB} = 0$$

$$F_{AB} = 40 \text{ kN (Tension)}$$



Solutions to Assignment #2, cont'd:

Average Normal Stress (definition: $\sigma_{\text{avg}} = \frac{N}{A}$):

The cross-sectional area of each of the bars is $A = \frac{\pi}{4}(0.02)^2 = 3.142 \times 10^{-4} \text{ m}^2$.

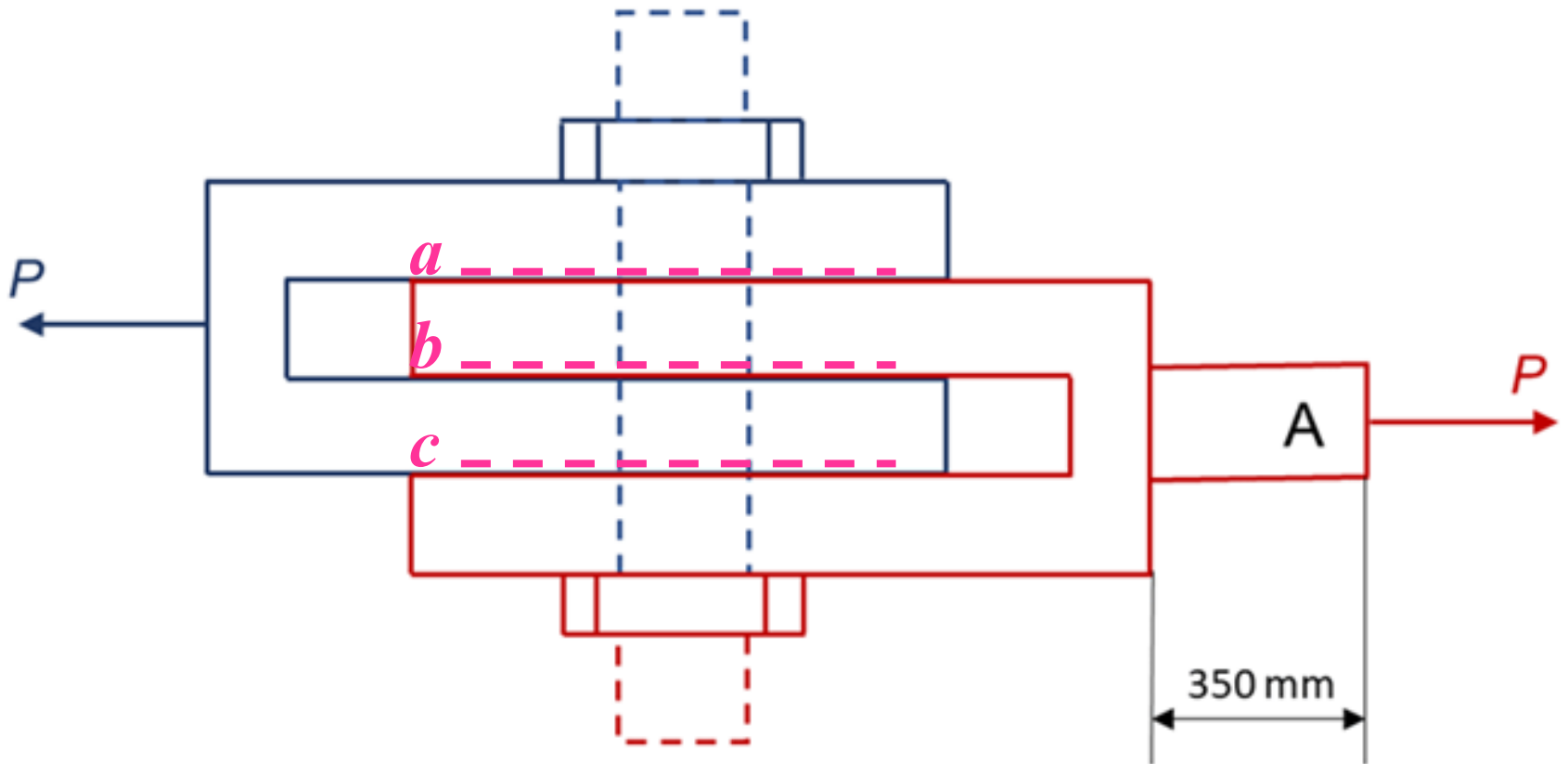
We can obtain the average normal stress

$$(\sigma_{\text{avg}})_{BC} = \frac{F_{BC}}{A} = \frac{50 \times 10^3}{3.142 \times 10^{-4}} = 1.59 \times 10^8 \text{ Pa} = 159 \text{ Mpa} \quad (\text{Ans.})$$

$$(\sigma_{\text{avg}})_{AC} = \frac{F_{AC}}{A} = \frac{30 \times 10^3}{3.142 \times 10^{-4}} = 9.55 \times 10^7 \text{ Pa} = 95.5 \text{ MPa} \quad (\text{Ans.})$$

$$(\sigma_{\text{avg}})_{AB} = \frac{F_{AB}}{A} = \frac{40 \times 10^3}{3.142 \times 10^{-4}} = 1.27 \times 10^8 \text{ Pa} = 127 \text{ MPa} \quad (\text{Ans.})$$

3. The pin is made of aluminum and has a diameter of $D = 60$ mm. Assume the maximum shear stress in the aluminum pin cannot exceed 200 MPa, determine the maximum external load P . (10' points)



Solutions to Assignment #3:

When the joints are subject to opposite load P , there are three (3) places / cross-sections in the pin that have relative movement (see the dashed lines). Using the equation of equilibrium for the left joint, for example, we get the relationship between the average shear force in each cross-section plane (V) and the external force (P)

$$P = 3V \quad (1)$$

Then, the average shear stress in each cross-section plane in the pin is

$$\tau_{\text{avr}} = V/A \quad (2)$$

If the joint is safe, the average shear stress in the pin should not exceed 200 MPa

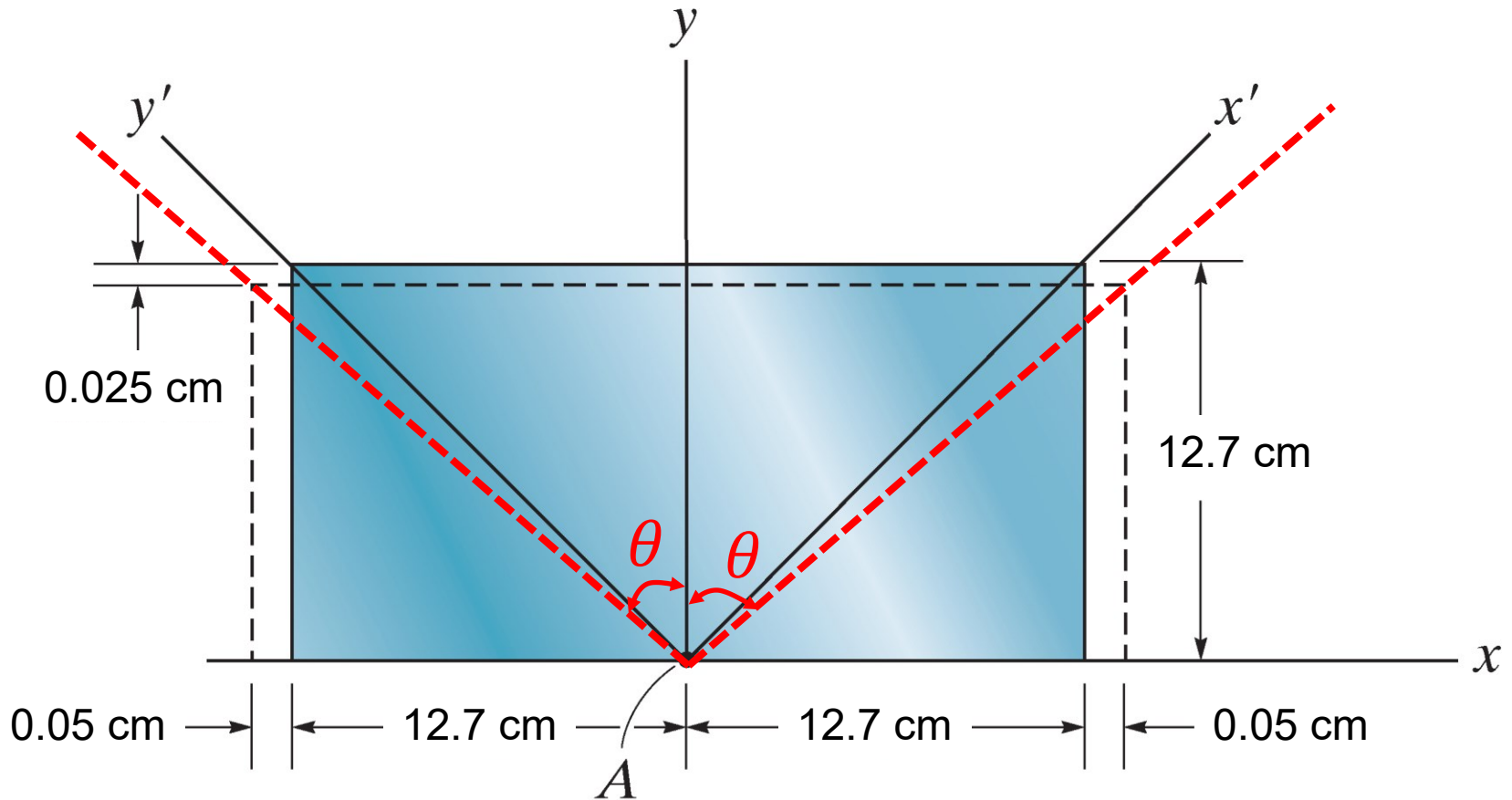
$$\tau_{\text{avr}} \leq 200 \text{ MPa} \quad (3)$$

Combining Eqs. (1), (2), and (3), we have

$$\frac{P}{3A} \leq 200 \text{ MPa}$$

$$P_{\text{max}} = 3 \times \frac{\pi}{4} \times 0.06^2 \times 200 \times 10^6 = 1696.5 \text{ kN} \quad (\text{Ans.})$$

4. The rectangular plate undergoes a deformation shown by the dashed lines. Determine the shear strain γ_{xy} and $\gamma_{x'y'}$ at point A. (10' points)



Solutions to Assignment #4:

The angle formed by Ax and Ay remains unchanged (90°) after deformation.

→ $\gamma_{xy} = 0$ (γ_{xy} defines **the change** in the angle formed by Ax and Ay) **(Ans.)**

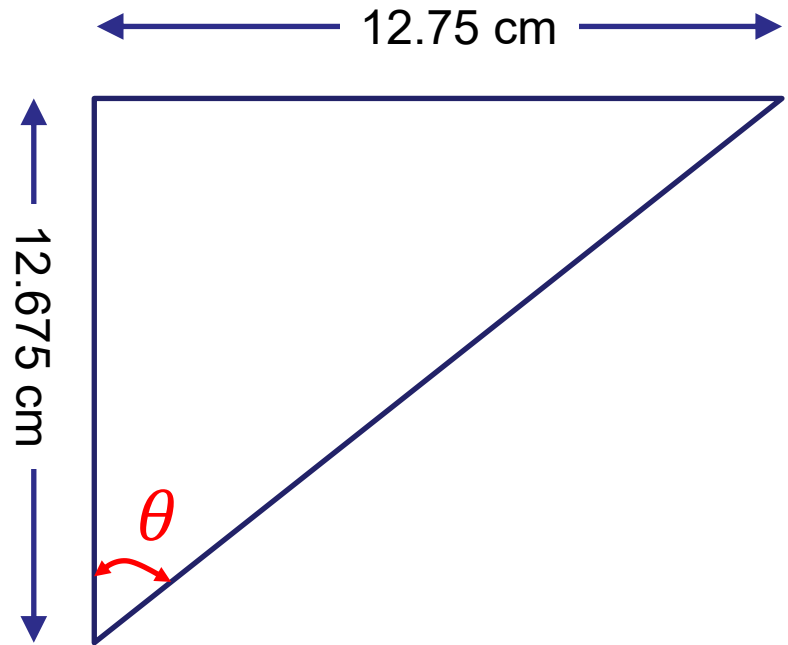
The angle formed by Ax' and Ay' has changed after deformation.

$$\tan \theta = \frac{12.75}{12.675}$$

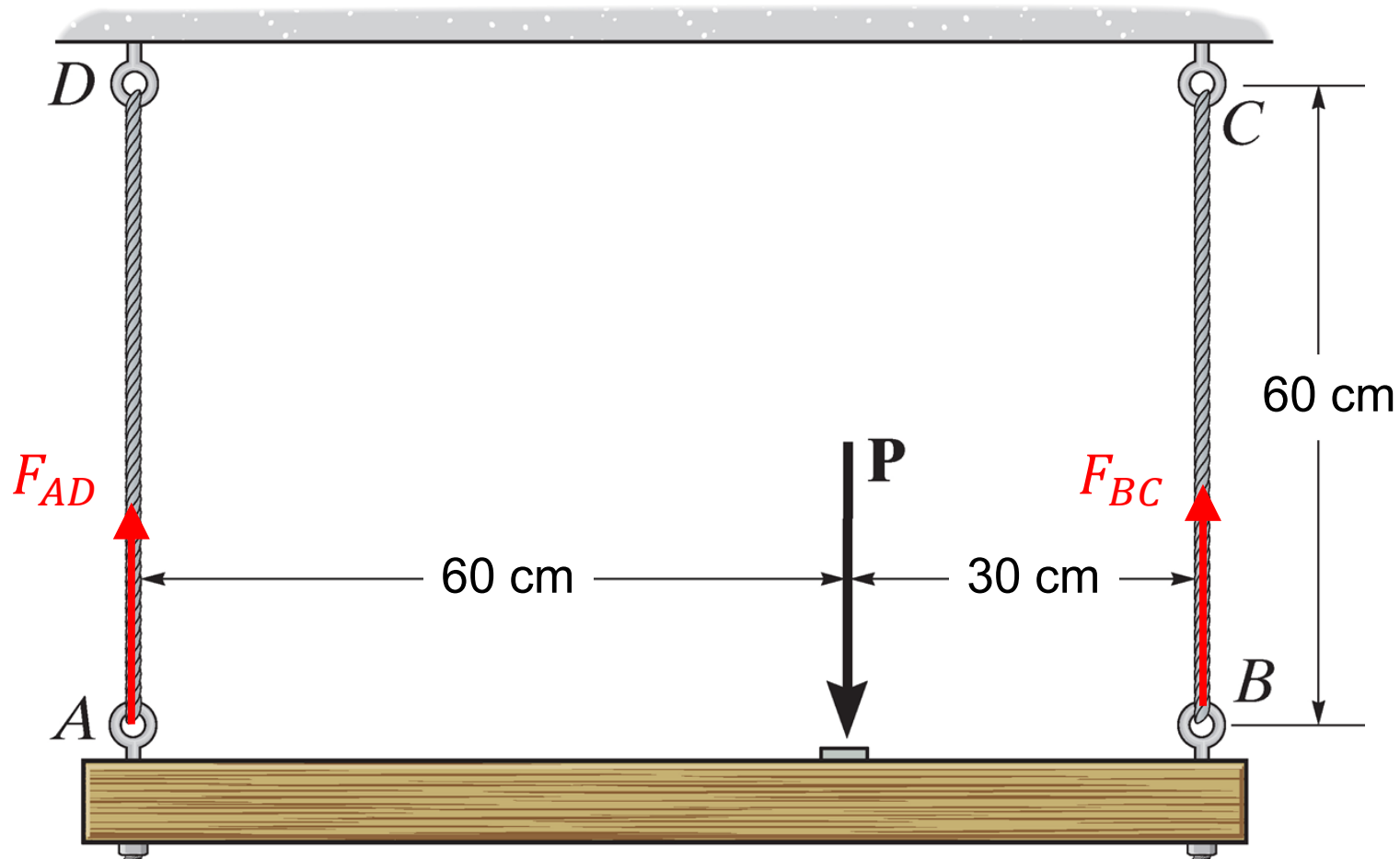
$$\rightarrow \theta = 0.78835 \text{ rad}$$

$$\rightarrow \gamma_{x'y'} = \frac{\pi}{2} - 2\theta = -0.00590 \text{ rad}$$

(Ans.)



5. The wires each have a diameter of 2 cm, length of 60 cm, and are made from 304 stainless steel (Modulus of Elasticity $E = 193 \text{ GPa}$). If $P = 30 \text{ kN}$, determine the angle of tilt of the rigid beam AB . Assume the deformation of the wires obeys the Hooke's law. (10' points)



Solutions to Assignment #5:

Equations of Equilibrium: Referring to the free-body diagram of beam AB shown

$$\begin{array}{lll} \sum M_A = 0 & 30 \times 0.6 - F_{BC} \times 0.9 = 0 & F_{BC} = 20 \text{ kN} \\ \sum M_B = 0 & 30 \times 0.3 - F_{AD} \times 0.9 = 0 & F_{AD} = 10 \text{ kN} \end{array}$$

Normal Stress in the wires:

$$\sigma_{BC} = \frac{F_{BC}}{A_{BC}} = \frac{20000}{\frac{\pi}{4} \times 0.02^2} = 63.6620 \times 10^6 \text{ Pa}$$

$$\sigma_{AD} = \frac{F_{AD}}{A_{AD}} = \frac{10000}{\frac{\pi}{4} \times 0.02^2} = 31.8310 \times 10^6 \text{ Pa}$$

Strain in the wires:

$$\varepsilon_{BC} = \frac{\sigma_{BC}}{E} = \frac{63.6620 \times 10^6}{193 \times 10^9} = 3.2985 \times 10^{-4} \text{ mm/mm}$$

$$\varepsilon_{AD} = \frac{\sigma_{AD}}{E} = \frac{31.8310 \times 10^6}{193 \times 10^9} = 1.6493 \times 10^{-4} \text{ mm/mm}$$

Solutions to Assignment #5, cont'd:

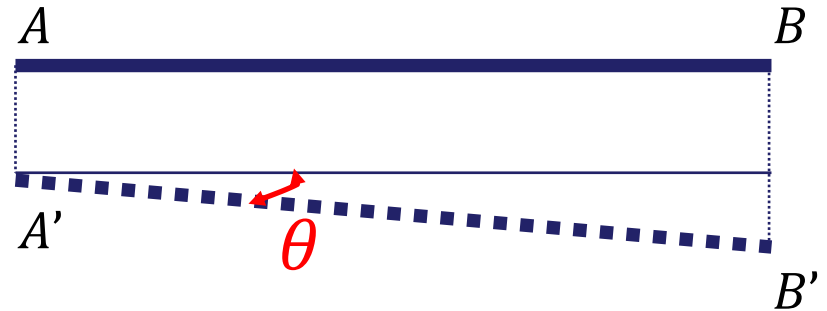
Elongation of the wires:

$$\delta_{BC} = \varepsilon_{BC} \cdot L_{BC} = 3.2985 \times 10^{-4} \times 0.6 = 1.9791 \times 10^{-4} \text{ m}$$

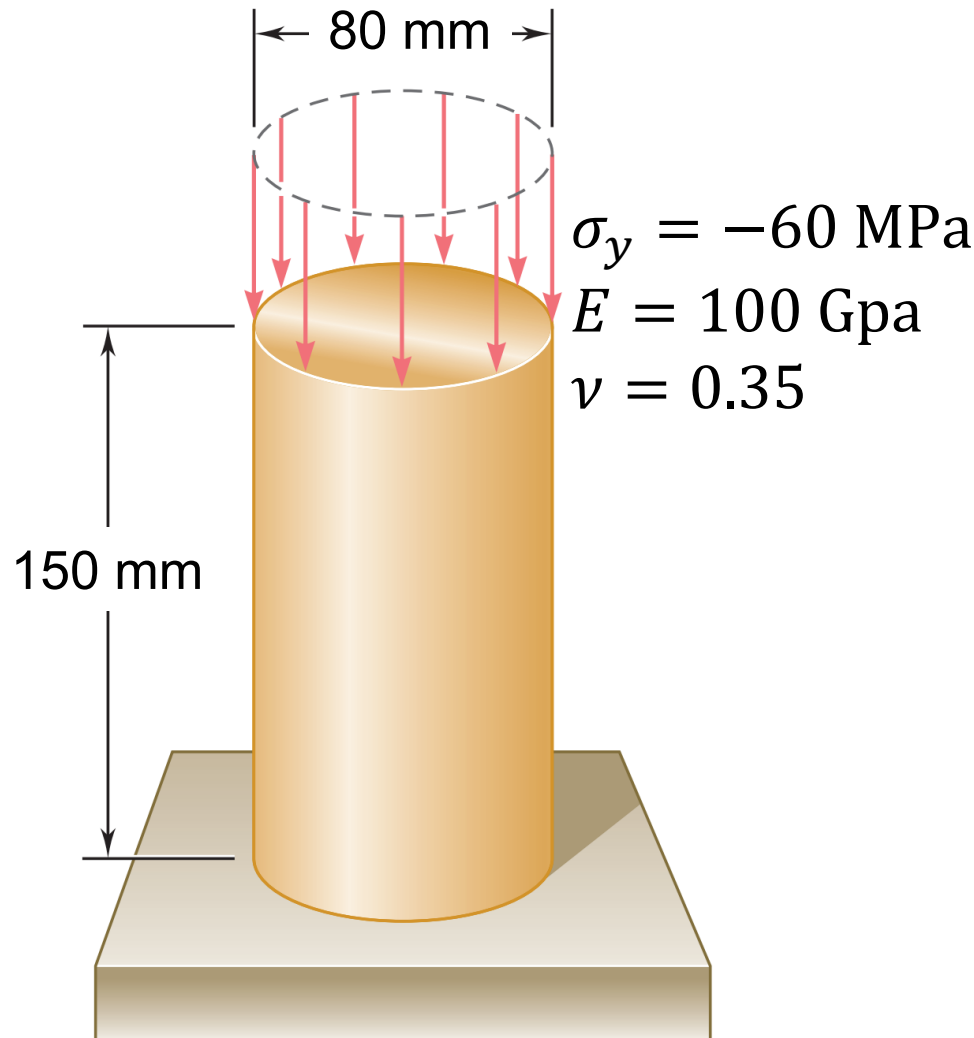
$$\delta_{AD} = \varepsilon_{AD} \cdot L_{AD} = 1.6493 \times 10^{-4} \times 0.6 = 0.98958 \times 10^{-4} \text{ m}$$

Referring to the geometry shown and using small angle analysis

$$\theta = \arctan \frac{(1.9791 - 0.98958) \times 10^{-4}}{0.9} = 0.006299 \text{ rad} = 0.4010^\circ \quad (\text{Ans.})$$



6. For the axial loading on a brass cylinder shown below, determine the change in volume of the brass cylinder. Assume the cylinder deforms in the elastic range. **(15' points)**



Solutions to Problem #6:

First, using Hooke's law, the longitudinal strain is (assume compression as negative strain and then expansion as positive strain):

$$\varepsilon_{\text{long}} = \frac{\sigma}{E} = \frac{-60 \times 10^6}{100 \times 10^9} = -6 \times 10^{-4} \text{ mm/mm}$$

Second, according to the definition of Poisson's ratio, the lateral strain is:

$$\varepsilon_{\text{lat}} = -\nu \cdot \varepsilon_{\text{long}} = -0.35 \times (-6 \times 10^{-4}) = 2.1 \times 10^{-4} \text{ mm/mm}$$

Third, according to the definition of strain, we can get the new dimensions (new height H and new diameter D) of the cylinder:

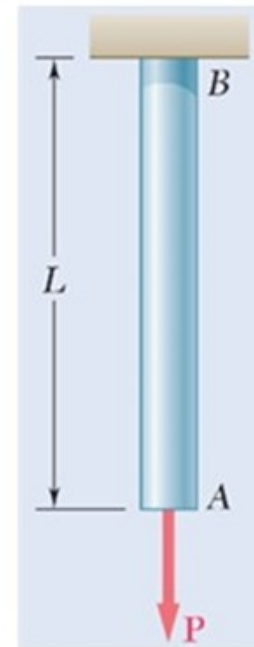
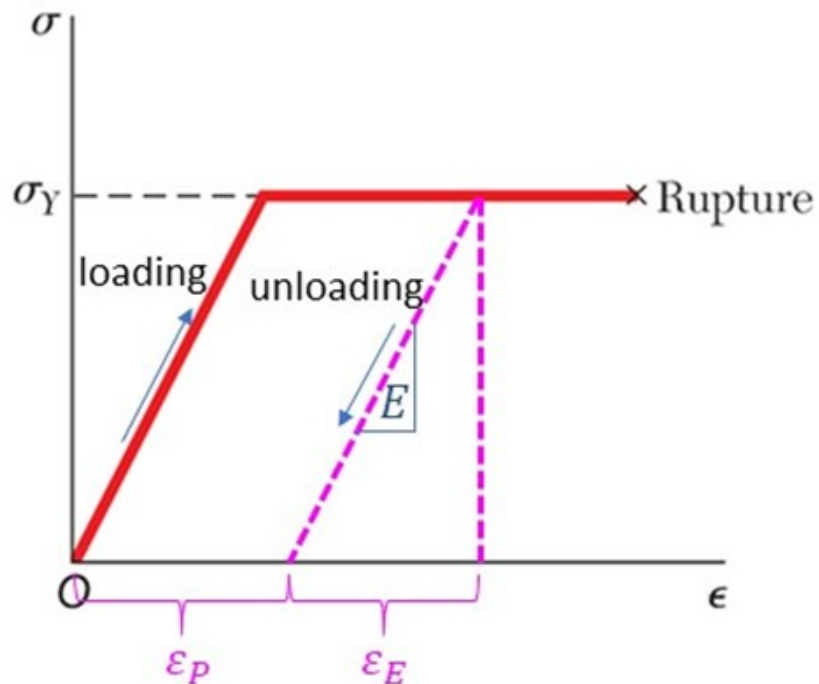
$$H = (1 + \varepsilon_{\text{long}})H_0 = (1 - 6 \times 10^{-4}) \times 0.15 = 0.14991 \text{ m}$$

$$D = (1 + \varepsilon_{\text{lat}})D_0 = (1 + 2.1 \times 10^{-4}) \times 0.08 = 0.0800168 \text{ m}$$

$$\text{Finally, the new volume is: } V = \frac{\pi}{4} D^2 H = 7.539822 \times 10^{-4} \text{ m}^3$$

Thus, the volume change is: $\Delta V = V - V_0 = -1.358 \times 10^{-7} \text{ m}^3 = -135.8 \text{ mm}^3$ (Ans.)
(The negative sign indicates the cylinder shrinks upon compression)

7. A cylindrical rod AB has a length $L = 80$ cm and a diameter of $D = 2$ cm. The modulus of elasticity is $E = 200$ GPa and yield strength is $\sigma_Y = 300$ MPa. The stress-strain diagram of the rod is shown in the left figure. Note that, after yielding the diagram is a perfect horizontal line. A force P is applied to the bar and then fully removed. This results in a permanent set $\delta_P = 0.2$ cm in the rod (right figure). Determine the value of the force P and the total amount δ_m by which the bar should be stretched. (20' points)



Solutions to Problem #7:

The permanent set is the plastic deformation in the bar ($\delta_p = 0.2$ cm), i.e. the total strain must be in the plastic deformation range. So, the external force must induce an internal stress that exceeds the yield strength. The minimum external force should be:

$$P = \sigma_Y \cdot A = \frac{\pi}{4} D^2 \sigma_Y = \frac{3.1415926}{4} \times 0.02^2 \times 300 \times 10^6 = 9.4248 \times 10^4 \text{ N} \quad (\text{Ans.})$$

We can draw the loading/unloading process in the stress-strain diagram.

The plastic strain part can be calculated based on the permanent set as

$$\varepsilon_{pl} = \frac{\delta_p}{L} = \frac{0.2}{80} = 0.0025 \text{ mm/mm}$$

The elastic strain part can be calculated by the Hooke's law (unloading) as

$$\varepsilon_{el} = \frac{\sigma_Y}{E} = \frac{300 \times 10^6}{200 \times 10^9} = 0.0015 \text{ mm/mm}$$

So, the total strain is $\varepsilon_{tot} = \varepsilon_{pl} + \varepsilon_{el} = 0.004$ mm/mm. The total amount of stretched distance is

$$\delta_m = \varepsilon_{tot} \cdot L_0 = 0.004 \times 0.8 = 0.0032 \text{ m} = 0.32 \text{ cm} \quad (\text{Ans.})$$